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Rank abundance of lady beetle population sampled from urban conglomerate of Gaya District, Bihar, India

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Abstract: Determination of rank abundance is one of the authentic statistical tools for quantification of species diversity of any biotic population. It provides firsthand information through mathematical and graphical means regarding the status of existence of multiple species in a population of any community. The present study is an effort to reflect the abundance status of lady beetle species existing in an urban conglomerate of western part of Gaya. As per the statistical data regarding relative abundance of six different species of lady beetles sampled from this area, the following rank abundance has been procure:- *Rodolia amabilis* (Kapur) > *Bothrocalvia lewisii* (Crotch) > *Hormonia axyridis* (Pallas) > *Coelophora bissellata* (Mulsant) > *Anisolemnia dilatata* (Fabricius) > *Coccinella septempunctata* (Linn.)

Keywords : Rank abundance, cumulative dominance, abundance status, ladybeetles species.

INTRODUCTION

Descriptions of whole communities by one statistic of diversity run the risk of losing much valuable information. A more complete picture of the distribution of species abundances in a community is gained by plotting proportional abundance (usually on a logarithmic scale) against rank abundance. The abundance of the most common species appears on the extreme left and that of the rarest species on the extreme right. A rank abundance diagram can be drawn for the number of individuals, biomass, ground area covered and other variables, all plotted against rank abundance[1].

There are at least 18 different theoretical forms of rank abundance diagrams, but only three most well known forms - the geometric series, lognormal, and broken-stick are of significant value. The distribution of species abundance is most equitable in the broken-stick, less equitable in the lognormal and less equitable still in the geometric series[2].

A common biological explanation for the pattern appearing in a geometric series is that the first or most dominant species to colonize a new area appropriates a fraction of the available resources and, by competitive interaction, preempts that fraction, say, 50%. The second species then preempts a similar fraction of the remaining resource, and so on with further colonists. Fits to this model have been found for plants from a subalpine forest community[3]. The model may best fit communities of relatively few species wherein a single environmental factor pre-dominates. Only one species is best fit to survive in such a habitat, so that species becomes numerically dominant. Both extreme temperatures (subalpine) and pollutants limit species richness—hence the fit of such communities to the geometric series might then be used as an indicator of a polluted community.

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The geometric series distribution contrasts starkly with the more equitable distribution represented by the "broken stick" model of Robert MacArthur[4] . In this model, the resources (represented by a stick) are divided at one time into segments over the whole length of the stick, implying instantaneous colonization by all species. The segments are ranked in decreasing order of their lengths. The abundance of each species is assumed to be proportional to the length of the segment associated with that species. A few bird, fish, worm, and predatory gastropod communities were found to fit the broken-stick model, but the model is realistic on relatively few occasions (Hairston, 1969).

The lognormal distribution is more common than the broken stick for most communities that are rich in species. One ecological explanation for the lognormal distribution refers to the broken-stick model, but the stick is broken sequentially, not instantaneously. The stick is broken at random into two parts, and then any one part is chosen at random and broken again, giving three parts. One of the three is chosen at random and broken again, and so on.

The lognormal distribution owes its place in ecology to Frank Preston[5], who obtained a normal, or bell-shaped, curve when he plotted number of species (on the y-axis) against the logarithm of species abundance (expressed as abundance classes [e.g., 1-10, 11-100, 101-1,000, and 1,001-10,000]) on the x-axis.

MATERIALS & METHODS

In the present investigation, sampling was carried out in the quadrats of Zone II being the South-West sites of Gaya, covering Paharganj, Cantt. area etc.

The land vegetation of the area has been also recorded for the description of physiography of the sampling sites through quadrat method. Zigzag area lying beyond the quadrats were ignored.

Sampling Methods : Since lady bird beetles are basically of visible size with brilliant colours and less motility, careful hand picking with the help of glubs and brush were practised as the sampling method. However, All Out Search (AOS) method had also been practised to collect all the representatives of as many as species of lady bird beetles as possible showing visible morphological variations[6,7].

Statistical analysis of Rank Abundance curve: The rank abundance of six ladybeetle species sampled in this study was statistically calculated as per stiling 2002[1].

Table showing sampling data, Relative abundance and Rank abundance of Lady beetles from western part of urban area of Gaya district (2016)

Sl. No.	Scientific names of the Lady beetle species	No. of individuals collected monthly			Mean Value	Relative abundance RA= n/N x 100	Rank Abundance
		July	Aug	Sep			
1.	*Rodolia amabilis* (Kapur)	24	39	43	35.33	24.3	**I**
2.	*Bothrocalvia lewisii* (Crotch)	12	27	37	25.33	17.3	**II**
3.	*Hormonia axyridis* (Pallas)	13	23	30	22	15.97	**III**
4.	*Coelophora bissellata* (Mulsant)	12	22	36	23.33	15.27	**IV**
5.	*Anisolemnia dilatata* (Fabricius)	13	20	27	20	13.88	**V**
6.	*Coccinella septempunctata* (Linn.)	14	19	24	19	13.19	**VI**
	Total	88	150	197	144.99		

Fig 1:- Monthly vartiation in the R.A % and related Rank Abundance status of six species of lady beetles sampled from western part of urban area of Gaya district.(2016)

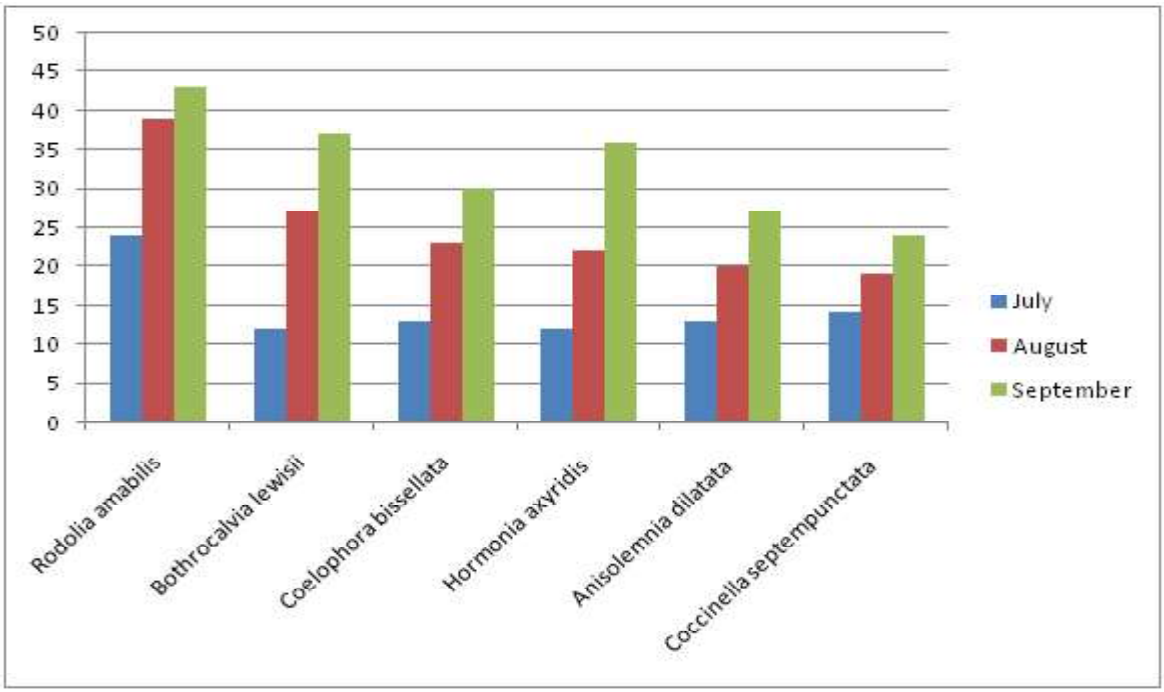
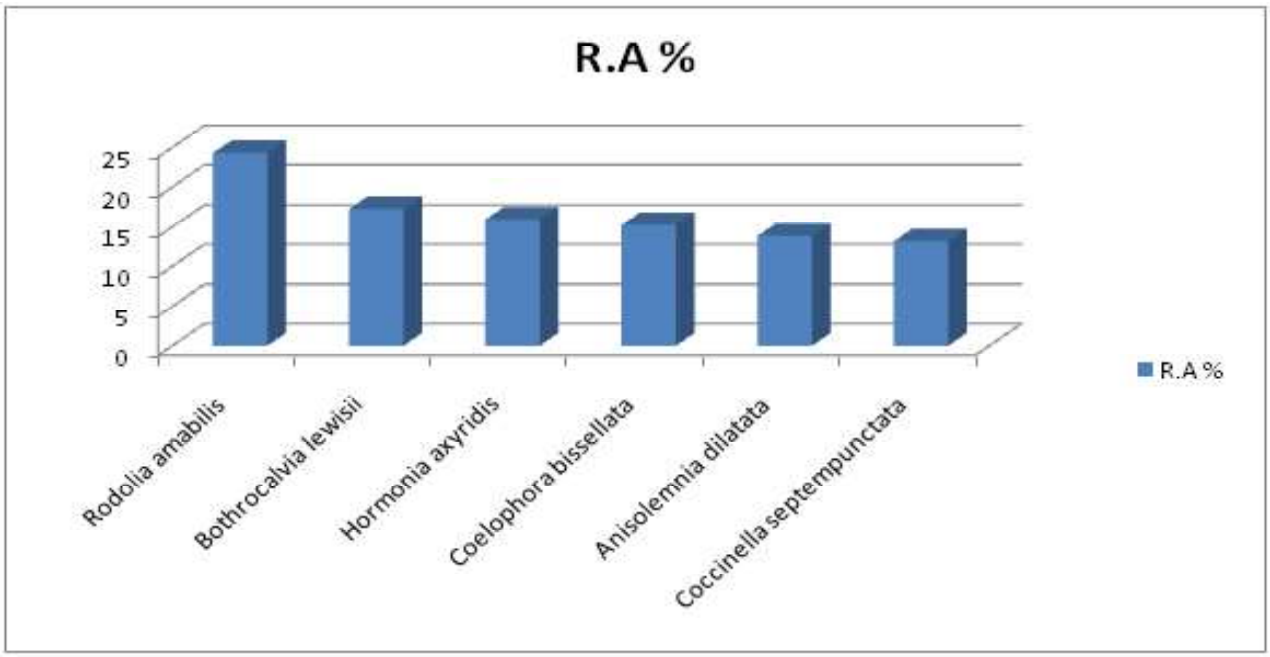


Fig 2:- R.A % dependent Rank abundance bar diagram of six species of lady beetles sampled from western part of urban area of Gaya district (2016)



RESULTS & DISCUSSION

Six different species of ladybeetles have been sampled from western part of Gaya city (urban conglomerate), a total of six varieties of ladybeetles have been randomly sampled during three selected months of July, August, September- 2016. These species are *Anisolemnia dilatata*, *Bothrocalvia lewisii*, *Coccinella septempunctata*, *Coelophora bissellata*, *Hormonia axyridis*, *Rodolia amabilis* .

Rodolia amabilis reflected maximum abundance in all the three months of sampling in the year 2016 hence occupied the first position in the rank abundance curve (Fig-2). Hence, it can be easily rated as the most dominant species supported by their prefered and favourable host plants. It contributed maximum proportion in the cumulative dominance of all the species in this zone. However, in 2016 sampling, it was again interesting to observe that another species; *Hormonia arvridis* showed maximum abundance in November 2016 while minimum in May 2016. Thus, relative dominance of these two different species was also observed in two different years of sampling in this zone.[4]

During the same sampling period, *Coccinella septempunctata* & *Anisolemnia dilatata* have been rated as the least & last but one dominant species occuping the sixth & fifth rank position respectively therewise contributing minimum proportion in the cumulative dominance of all the species in this zone. However, some deviated trends of individual dominance and rank abundance in the sampled species have also been recorded which did not affect the overall rank abundance curve and cumulative dominance.

This peculiar trend of rank abundance profile in relation to the individual and cumulative dominance of the species sampled from this zone clearly reflects their intrinsic relationship with the available vegetation and meteriological conditions as well as the interspecific strugle for existance. Such kind of findings related with studies on insect species diversity and its temporal and spatial variations in tropics may provide important clue to predict the phylogenetic status of the concerned species and their role in the maintenance of the environmental equilibrium for safeguarding the vegetational species[7,8].

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